Raquel Dickinson Field Camp Day 8: ~~[illegible]~~ NS Seismic 24 ~~[illegible]~~ MAY 2026

Group: Group Lead - Chandler Data Lead - Raquel TA - Julia
Safety Lead - Alex Other - Samantha

Safety:

Safety Lead: Alex

Emergency Communication Channels: Raquel's #: (908) 210-1897
Emergency Contact (Betty Dickinson): (908) 295-7313
Radio Channel: 6
Emergency: 911
Matt: (317) 671-3437
Brandon: (713) 703-1289
TA #: (425) 289-6138

Site-Specific Hazards: There are people around! Be careful not to swing hammer with people around.

Equipment-Related Safety Protocol: Stay to the side of the hammer (not in front or behind), check hammer head, hard hats, gloves, its loud → ear protection.

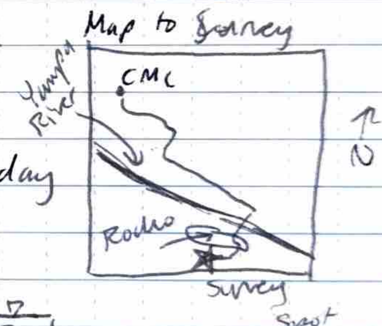
Site Information:

Site Name: Howelsen Park

nice!

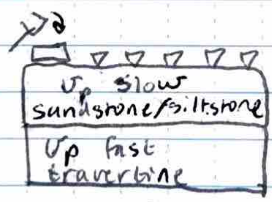
GPS: 40°28'52" N, 106°50'5" W

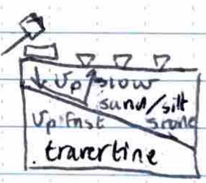
Site Conditions: Sunny, occasional drizzles, cloudy later in the day



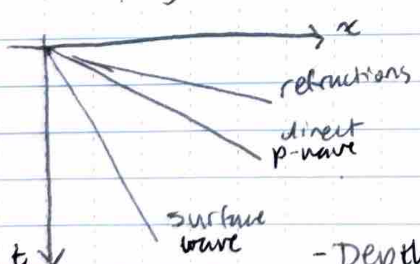
Background:

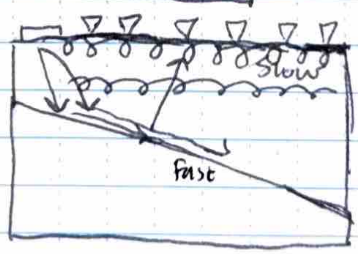
- Travertine deposits
- Water velocities: ~1500 m/s
- Air/sound velocities: 340 m/s
- Looking at p-wave velocities





← likely a dipping layer





surface waves are slower than body waves

- Depth to travertine is probably on the order of ~~[illegible]~~ 10 m.
- Browns Park Formation

great notes!!

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Method Information (Cont.)

Equipment:

- hammer
- extention cable
- battery ×4
- geode ×2
- battery-geode cable
- takeout cable ×2
- geophones ×48
- computer ×1
- yellow data cable ×2
- off-road cart
- hard hats
- GPS → base station → emlids/rover

Survey:

Parameters (preliminary): We used 2m spacing and a line length of ~~96~~ 94m, not 100m. We plan to use $f_s = 500$ Hz, 4 stacks, 2 s listening time, and we will hit at ~~every other node.~~ at every 3rd node.

Equipment: Double ethernet → use P1 not P2 great note

Test Shots:

Trial:
- Make sure the trigger is on top when swinging
- Green bar → ready to hit
- Yellow bar → triggered
- Coupling at 0m is poor due to sticks
- Stack 1: Not much noise → 4 stacks for survey

Shots start at 1320

Shot ~~Notes~~ Notes:

love the table! maybe take a pic and upload to the data onedrive

	File:	Distance(m):	Coordinate / Distance ~~Shot #~~:	Notes:
1.	3.DAT	0	0	Alex Swing
2.	4.DAT	6	6	
3.	5.DAT	12	12	
4.	6.DAT	18	18	
5.	7.DAT	24	24	
6	8.DAT	30	30	
	9.DAT	36	36	
	10.DAT	42	42	~~Chandler~~ Sam Swing
	11.DAT	48	48	Stack 1 → radio fell
	12.DAT	54	54	plate on the grass, rain started
	13.DAT	60	60	
	14.DAT	66	66	Problems w/computer
	15.DAT	72	72	Chandler, some movement b4 hits
	16.DAT	78	78	Grass plate
	17.DAT	84	84	plate on grass
	18.DAT	90	90	plate on grass
	~~19.DAT~~	~~96, 94~~	~~96, 94~~	

GPS is taken at 94m and 0m : 94m: 13T 0344495 UTM 4482781

0m: 13T 0344505 UTM 4482779

Scale: 1 square = ________

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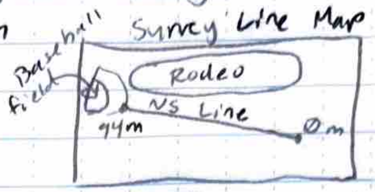
Survey (Cont.)

~~Shot Notes (Cont.)~~

~~File: Distance (m): Coordinates: Notes:~~

Parameters: $f_s = 500\,Hz$, $T_s = \frac{1}{500}s$, stacks = 4, spacing = 2m, listening time = 2s, line length = ~~100~~ 90m, hammer shot density = every 3rd node = 6m

Survey Line Map



Weather: It was mostly warm and sunny, but at 1218, it began to drizzle, but only for a short while. We ~~covert~~ covered the computer in a rain jacket to protect it.

Data Quality Factors: The line length may not have been long enough to get the most optimal ~~reflection~~ refracted waves. We also saw that channel 18 had a lot of noise, so that node should be disregarded. There was noise from moving before shots, and there were a few "double shots" where the hammer likely bounced. People were walking and riding their bikes, but we waited for them to be gone before recording.

Data Collection & Storage:

* 7.DAT and 8.DAT → normalize

* Channels 32 on look flat (9.DAT)

0.08 s at ch 33, ends at 0.17s → $\frac{30m}{0.09s} \approx 333.33\,m/s$ → speed of sound in air ← mute!

0.06 s at ch 38, ends at 0.07 s at ch 48 → $\frac{20m}{0.01s} = 2000\,m/s$ → speed of sandstone?

Drier at top of hill → worse coupling at early channels → noise

File 16.DAT: velocity ~3400m/s (more zoom → ~~more~~ easier time estimates)

Zoomed → $v \approx 4300\,m/s$ → traveltime?

Channel 18 → no good.

Where: One Drive : gpgn486-2026data > near_surface_seismic > 2026_05_24_team3_ns_seismic

Format: .dat and .png

Header File: N/A for now

Other datasets: N/A gps too!

Data Lead: Raquel

Scale: 1 square =________

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Wrap-Up:

Safety: Concerns: - carrying heavy equipment uphill
- slips, trips & falls
- sun → water & sunscreen
- tall grass → bugs & snakes → gloves & look!

Near misses/Incidents: N/A

Tasks Accomplished: ① Completed 90m survey with usable data
② Have started to see a high velocity anomaly → travertine?
③ Stayed safe!

Challenges: ① Terrain had limited space for the survey
② Lots of people and bikes → noise & safety hazard
③ Computer bugged out → restarted software

Future Work: ① GPR survey of area → "confirm" travertine
② Longer survey line with tighter spacing (NS seismic)